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Smart Distribution Systems

Exercise session 2

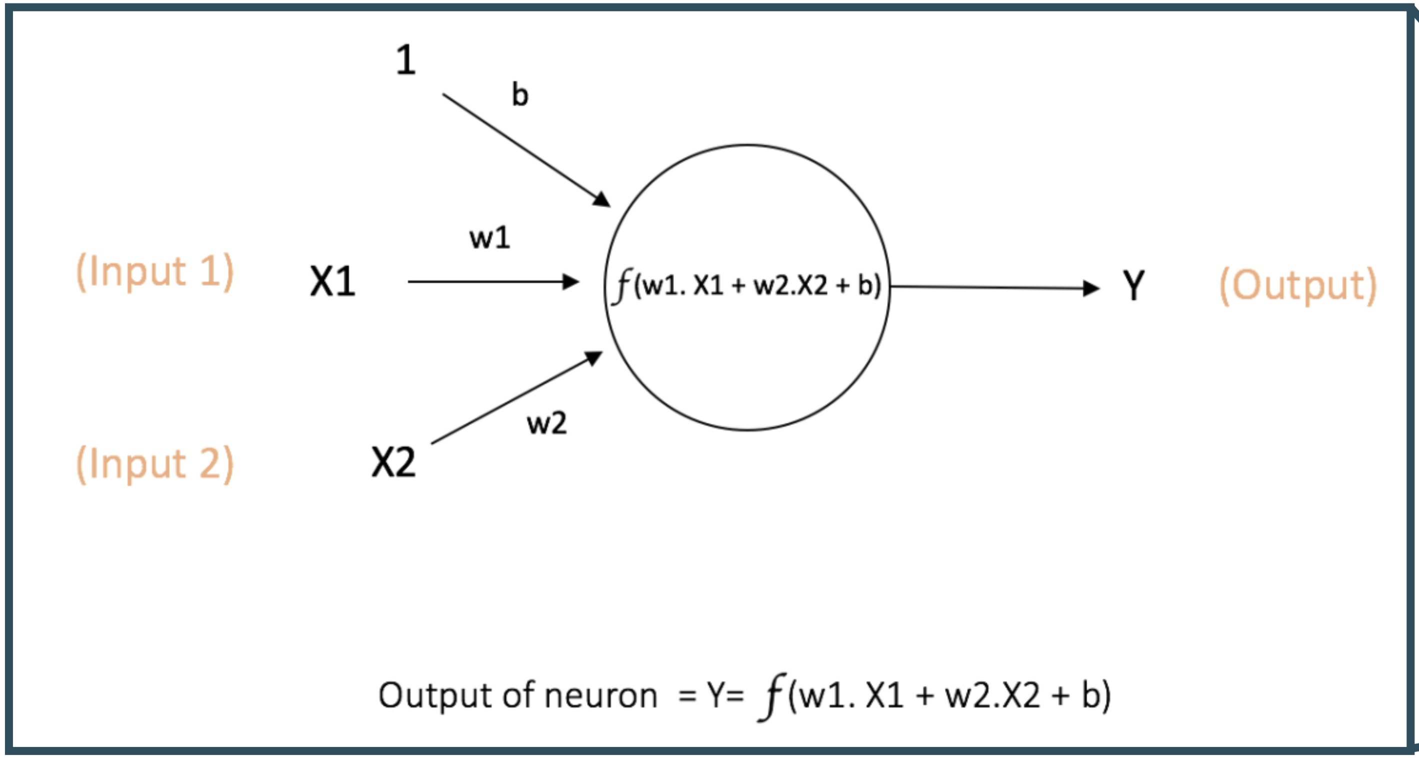
Sina Toghranegar– sina.toghranegar@kuleuven.be

Naj Masoudi– najmeh.masoudi@kuleuven.be

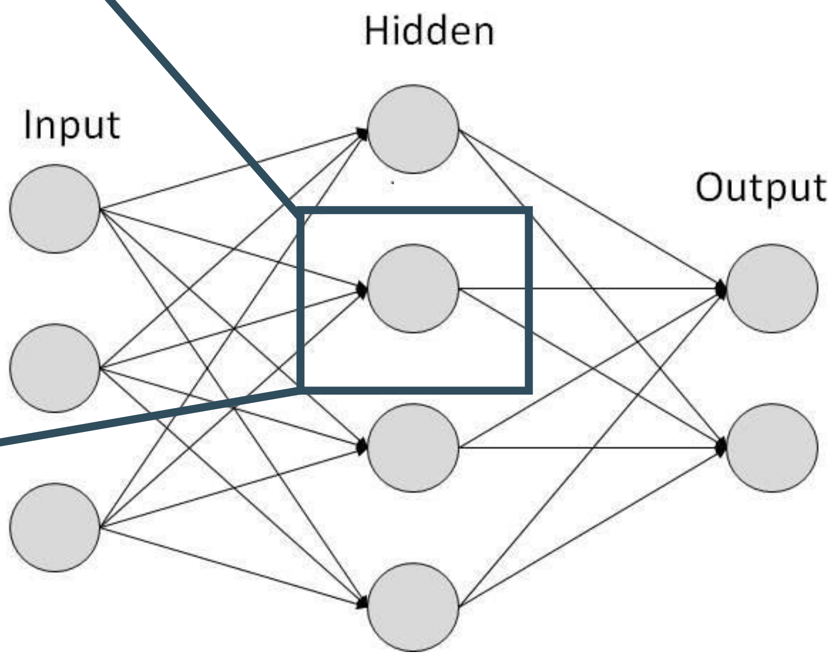
Outline

- Neural networks
- Training-, validation- and test-set
- Exploration
- Neural network architecture
- Feature selection

Neural networks



$$y = W\,\sigma(Vx + \beta)$$



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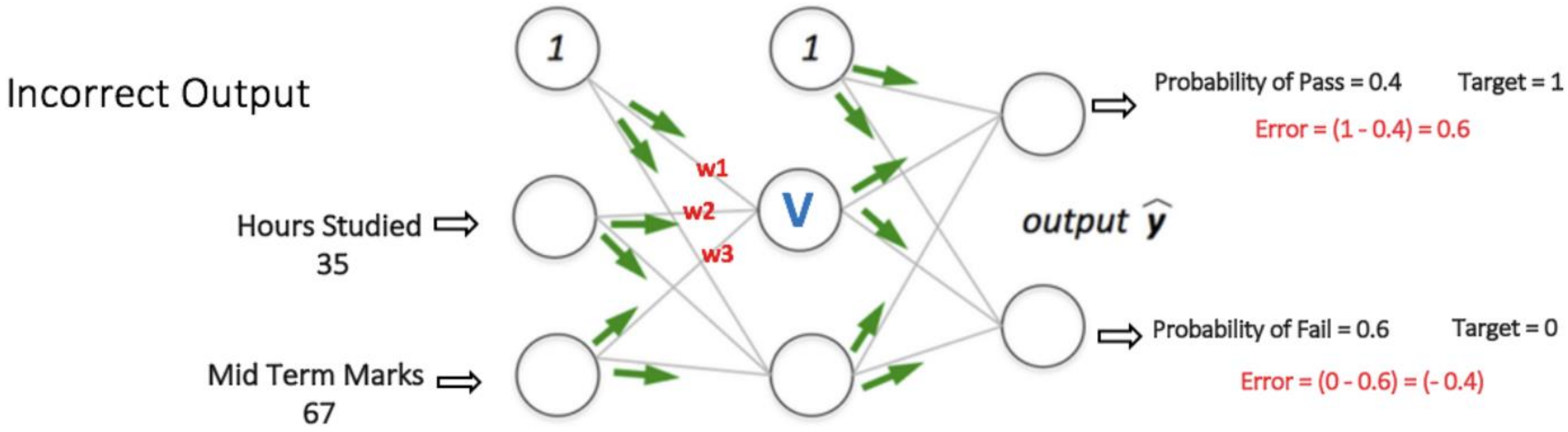
Neural networks – feedforward



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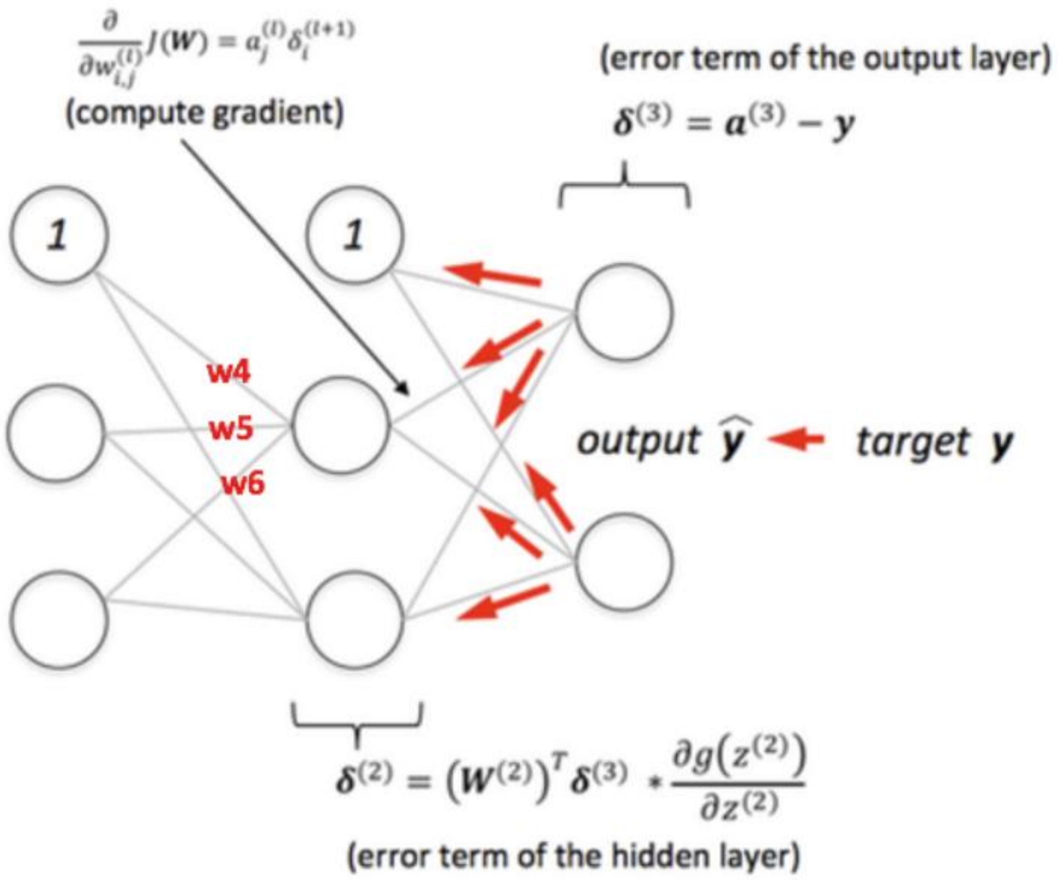
Neural networks - backpropagation

Backpropagation
+
Weights Adjusted

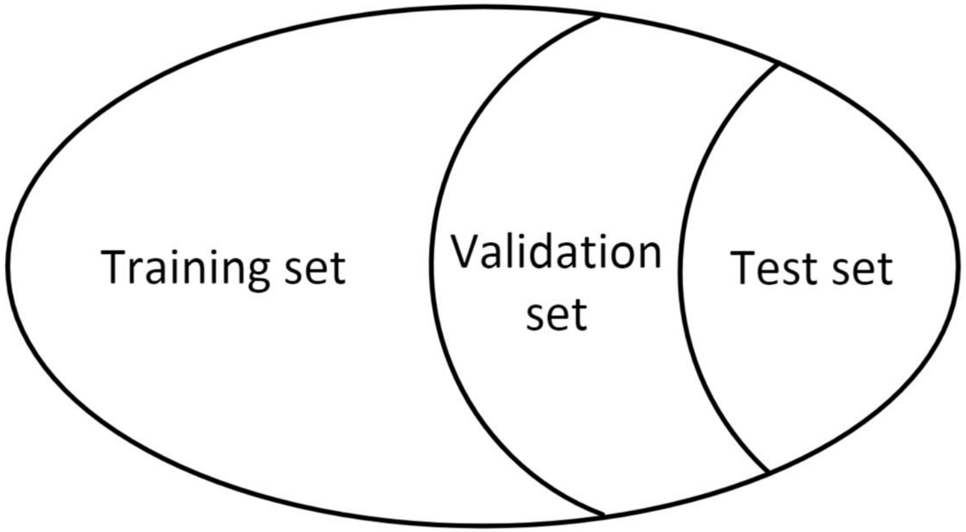


Gradient descent

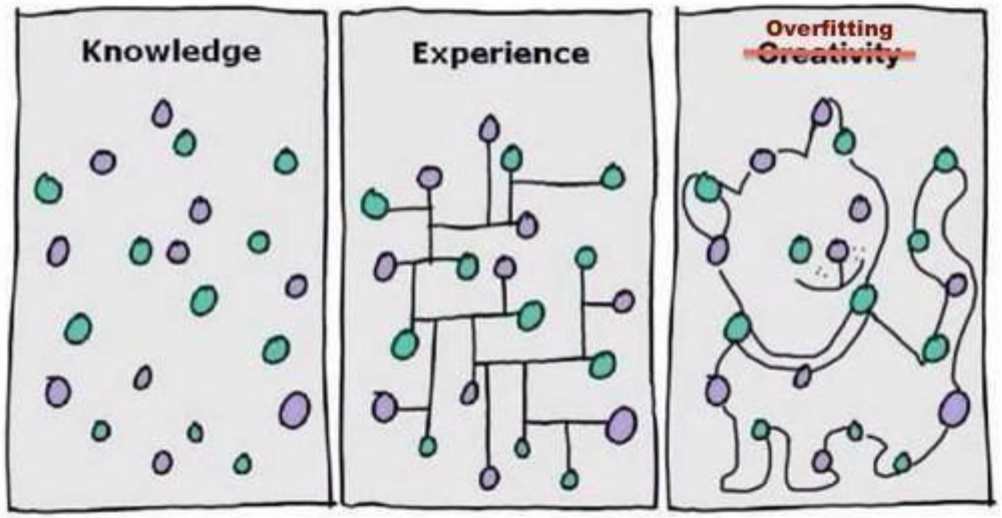
- This means we have a regular optimization problem in the weight-space
    - Solve with gradient descent

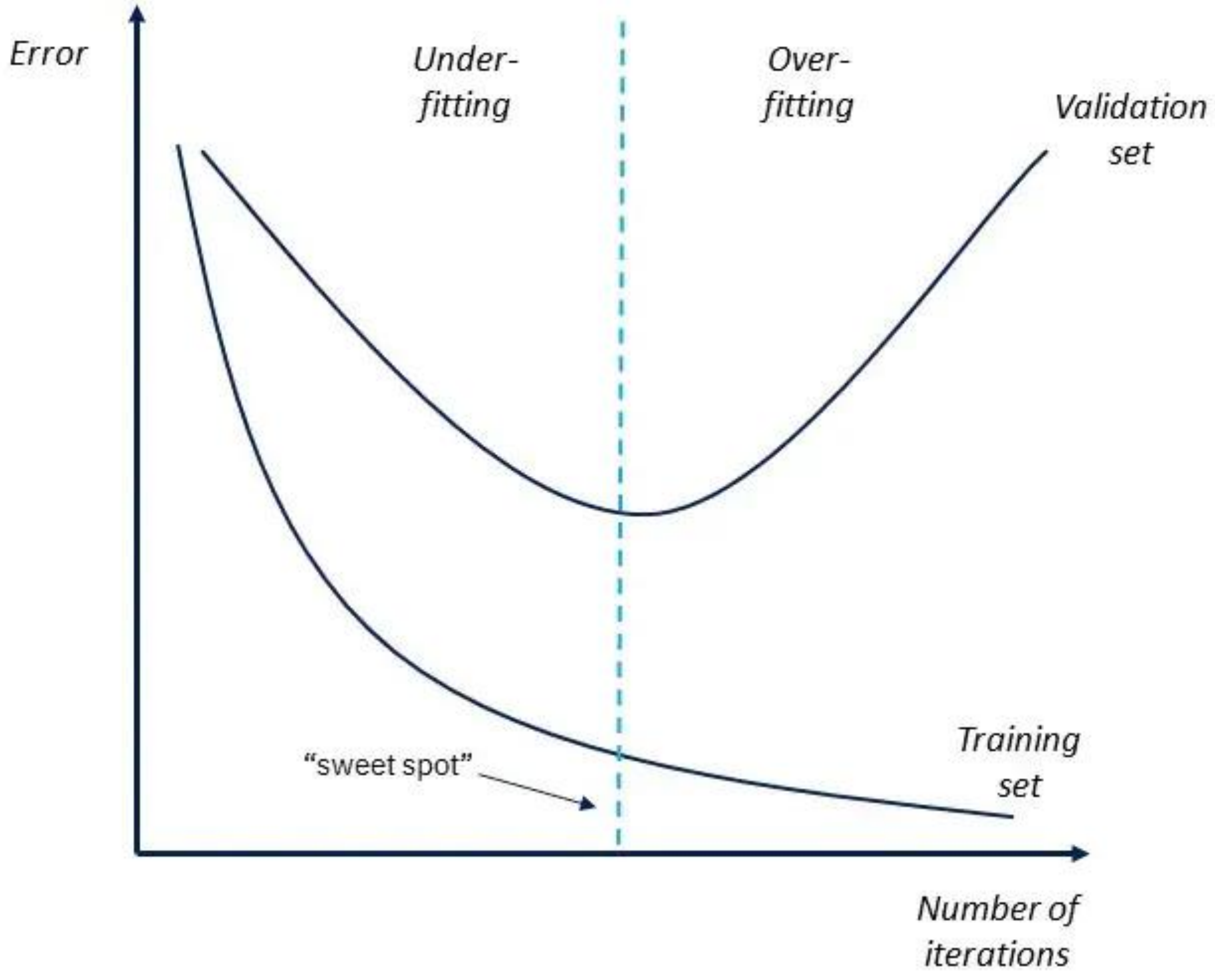
Training-, validation & test-set

- Training set: learning set of nn parameters
- Validation set: optimizing nn hyperparameters
- Test set: evaluate nn performance

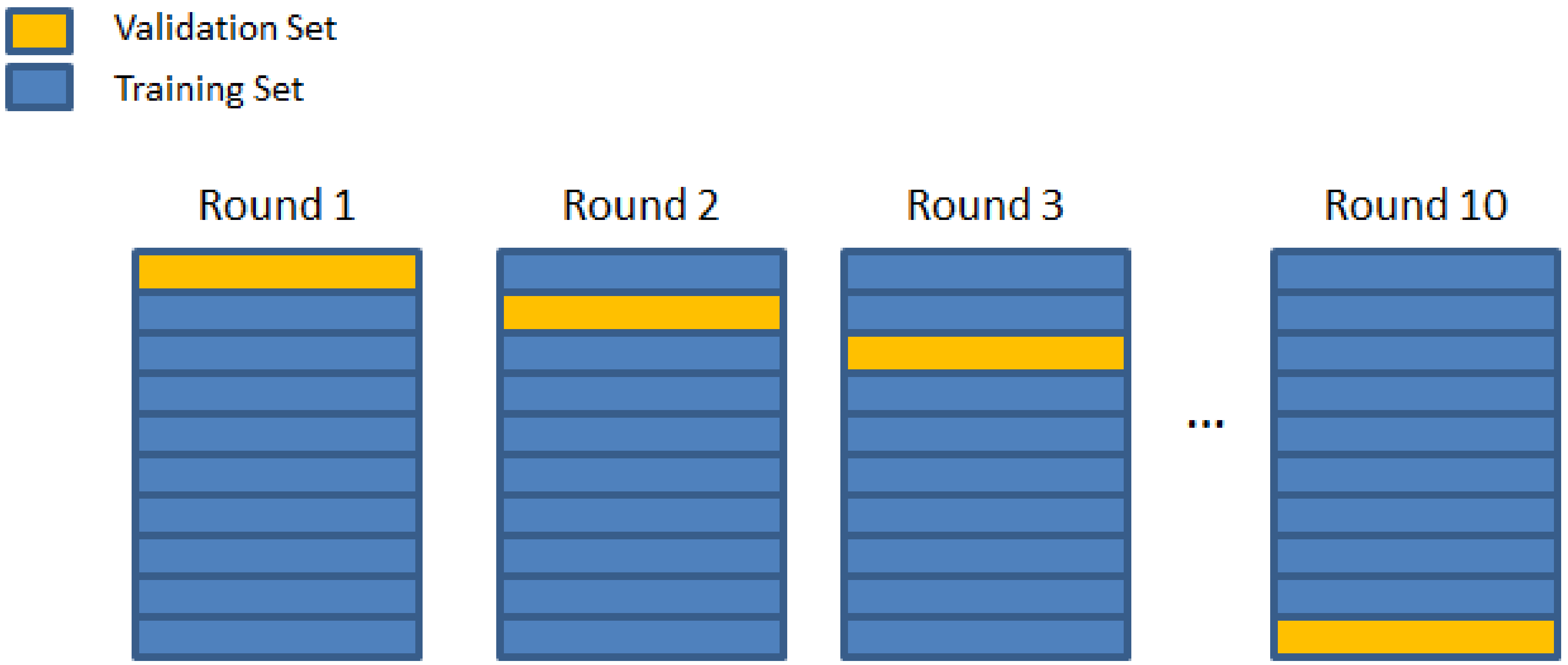


Overfitting





Cross-validation



Data set exploration

- First! Explore problem and data sets
  - What do we need to forecast?
  - Which factors influence the target?
    - Do we have data for them?
  - What does the data look like?
    - Is there seasonality?
    - Is there a trend?
    - How about outliers?

Neural network architecture

- Which architectures are we going to consider?
  - How many hidden layers? How many neurons?
  - Which activation function performs best?
  - How many outputs?
- Test based on validation set
- For example: 24 nn’s for 24 forecasts, or 24 outputs?
- Test multiple times!
  - Performance depends on (random) initialization.

Feature selection

- Which features can influence the performance?
  - Day of the week
  - Previous prices
    - How many previous hours?
    - Maybe only same day of previous week?
    - Current running average?
  - Solar and wind production?
  - Day-ahead prices of France, … ?
- Does normalization of features (and output) help?
- Selecting relevant features can make a big difference!